Requirements	Implications
1. Build two drones(one scout and one delivery) or one hybrid (scout + delivery) drone.	1. Two specialized drones, i.e. scout drone for scanning and delivery drone for deploying survival kits / One hybrid drone that performs both the functions.
2. Area coverage : scan 30 hectares area.	2. High endurance battery, efficient camera for scanning, efficient flight path planning.
3. Sending real time video	3. Onboard camera and a high bandwidth video transmitter(VTX) and a video receiver(VRX).
4. Geotagging survivors	4. Computer vision or object detection algorithms like YOLO, onboard GPS module to record the exact GPS co-ordinates of the survivor.
5. Audio communication with the survivors	5. Speaker module or buzzers to alert the survivors, which should be lightweight.
6. Deploy a delivery drone to deliver survival kits.	6. Inter-drone communication or centralized Ground control station that takes data from the scout drone and deploys the delivery drone.
7. Payloads of dimension (5 x 10 x 20 cm, 200g) attached to the delivery drone and their automated delivery.	7. A box like structure attached to the drone which carries the payload, the delivery mechanism can be facilitated by servo motors.
8. Lower the drone’s altitude to at least 20 feet and drop payloads near an actual person within 1.5 m radius.	8. Standard GPS is for horizontal mapping, so we need to add a downward facing sensor like a LiDAR or an ultrasonic sensor pointing towards the ground.
9. Return to home feature.	9. The drones must fly themselves from a single station. If the battery gets dangerously low or the radio signal drops, the drone must automatically fly back to that home station.
10. Weight & Size Limits: Max takeoff weight is 25 kg per drone, and they must fit in a 6 ft x 6 ft launch area.	10. Appropriate frame material selection and thrust-to-weight ratio calculations for the propulsion system, compact frame design or foldable arms.
11. Both drones must fly autonomously while reporting their status from a single command and control station.	11. Requires an advanced flight controller (e.g., Pixhawk running ArduPilot/PX4), multi-vehicle management software, and a robust telemetry link on a delicensed frequency.

12. Drones cannot fly above 400 feet. They must also have a geofence breach feature to automatically return home if they fly outside the area. 13. In case of a *critically* low battery, the drones must perform a gradual descent in the same location instead of trying to fly all the way home.	12. The flight controller firmware must be configured with strict virtual software boundaries (Geofence) and a hard altitude ceiling of 400 ft. 13. Configure the flight controller software with a two-stage battery fail-safe. Set a primary voltage limit to trigger return-to-home and a secondary, critical voltage limit that forces the drone to immediately perform a slow, controlled landing exactly where it is to prevent a crash.